Syringe handling

Two robot arms set doses, draw from a vial, inject and flush a saline lock at a skills-laboratory counter

Gym-Anything for Robotics, environment `syringe-handling`.



The bay as built and rendered: a clinical skills-laboratory counter with the tray, the kidney basin, the injection pad, the vial and the sharps container, and the two arms on their cart.

Environment `syringe-handling` of Gym-Anything for Robotics: eight tasks with syringes, a vial, an injection pad and a saline lock at the counter of a clinical skills laboratory, done by two FR3 arms. This report states the work, the scene, the physics of what a syringe holds and of its needles and fittings, the robot's runs, the benches that check each model, and what is not right.

The default policy camera is a software renderer that draws the cues the tasks need: where the gas is in a syringe, the liquid in a tilted vial, the injection ring, printed labels and scales, a wall clock that runs with simulated time, and deposits where liquid lands. On it, eleven HTTP commands across two sessions, a renderer restart at the published floor among them, ran at most 47.89 times real time, startup separate (section 18). A Unitree G1 with Dex3 hands has upper-body joint control and a contact-supported syringe carry; its full trial failed (section 23). Precision force calibration, transfer to a patient, free humanoid balance and full-task autonomy are not claimed.

01 / The work as it is done

Eight tasks a nurse or a pharmacy technician does many times a day, as they are taught.

The plunger is the last ten seconds of a longer piece of work. What makes it hard is what the plunger pushes against and everything that has to be right before it is pushed: the needle unsheathed without a touch, the air out, the dose on the scale, the connection made and turned, the clamp open, the needle in the right layer, the rate held, the sharp put away. The research read the procedures as they are taught (Doyle and McCutcheon, Clinical Procedures for Safer Patient Care; BD's flush-syringe guidelines) and the standards the equipment is made to (ISO 7886-1 syringes, ISO 7864 needles, ISO 594 Luer fittings, ISO 8362 vials), and chose the one place where all of it happens on a counter: a bay of a clinical skills laboratory, where students practise on an injection pad, a vial of coloured water and a training arm with a saline lock.

task	handed over	done when
set-dose	a sheathed 3 mL syringe holding 2.8 mL and 0.3 mL of air	2.5 mL within ISO 7886-1's tolerance, at most 0.05 mL of air, the excess in the basin
draw-dose	an empty 3 mL syringe with a blunt fill needle, a vial	2.5 mL drawn, no air, the vial upright where it stood and within 20 kPa of ambient
inject	a sheathed syringe holding 0.5 mL, the pad	90 % of the dose in the muscle, the needle in the ring and square, no faster than 1.5 times 10 s per mL, the syringe in the container
vial-to-pad	a syringe with a fill needle, a 25 G needle in its sheath, the vial, the pad	the draw, the needle change, the injection and the clearing away, each as above
sharps-away	three used syringes with bare needles on the tray	every syringe inside the container through the flap, the flap closed
flush-lock	a prefilled saline flush, the saline lock clamped	95 % of 5 mL from the flush syringe into the line (initial line saline excluded), the clamp closed and the connector bare at the end
slow-push	4 mL of medication in a labelled syringe, a flush	the dose in the vein over at least 86 s and never faster than 0.75 mL in 15 s, then the flush
fast-bolus	10 mL of viscous contrast, a flush	5 % to 95 % of the contrast out in under 7.2 s, the line under 300 kPa, then the flush

Seventeen variants change one thing each: a subcutaneous dose, a vented vial adapter, a half-empty vial, a vial left under vacuum, an upright upper-arm model, a blocked cannula, a clamp found open, two other cannula gauges, a one-minute push. Every task also holds eight conduct criteria: everything at rest at the end, nothing off the counter, no protective stop of an arm, at most 0.1 mL spilt, no bare needle or nozzle touching what it may not, no sheath pushed back on by hand, no needle loaded past its yield moment, no plunger pulled out or thread overridden.

02 / The bay at real scale

Objects at their makers' dimensions and masses; the room is dressing except where it is touched.



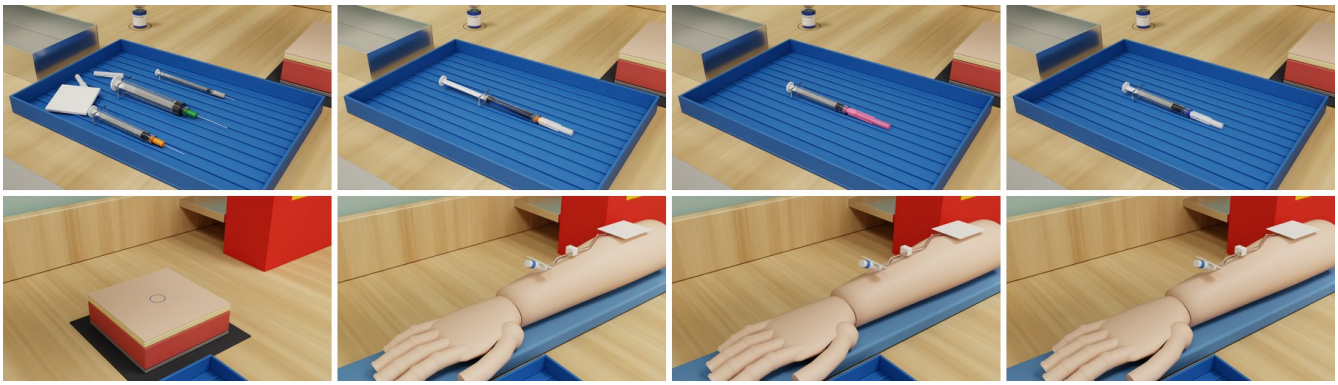
The bay from the room's far corner: the counter with the hand-wash sink, the two FR3 arms on their cart, the drip stand and the bed behind. Rendered from the simulated state, as every picture here is.

object	as built	basis
syringes	1, 3 and 10 mL Luer-lock and a 10 mL saline flush; bores 4.70, 8.59 and 14.43 mm; 3.4 g empty (3 mL)	makers' bore tables; ISO 7886-1 dead spaces
needles	18 G blunt fill, 21, 23, 25 and 27 G with hubs and sheaths in ISO 6009 colours; 0.25 g with its hub	ISO 7864, ISO 6009, Hamilton gauge chart
vial	10 mL glass with a 20 mm crimped stopper	ISO 8362-1
injection pad	130 mm square: skin 2 mm, fat 8 mm, muscle 30 mm over a board	trainer makers' figures; puncture from published needle studies
saline lock	needle-free connector on a taped mount, a line of 1.1 mL, a slide clamp, a 20 G cannula into a vein	BD flush guidelines; cannula makers' figures
sharps container	270 by 100 by 270 mm with a slot and a counterbalanced flap	container makers' figures
counter	beech, 0.914 m high, 0.635 m deep, a ribbed polypropylene tray and a steel kidney dish	ward furniture standards

The room is 3.6 by 3.0 m and 2.7 m high: sage wipe-clean cladding, wall trunking with gas outlets, a clock whose hands show the simulated time, six ceiling panels, a bed with a curtain and a drip stand. The walls, the floor, the counter, the sink and the trunking are colliders; the rest is dressing and is listed as such in `dressing.py`.

03 / Start states and what a policy sees

Every start state as a person leaves it: at rest for 60 s with the robot present before anything acts.



Start states of the eight tasks, close: the tray, the vial, the pad, the saline lock and the sharps container as each task hands them over.



What a policy sees in set-dose: the station camera on the wall, the overhead camera and the two wrist cameras.

A policy is given images from the cameras it asks for (`station_rgb`, `overhead_rgb`, a wrist camera on each hand, and `bay_rgb` from the ceiling behind it, the one camera that sees the wall clock slow-push's instruction times the push by), its arms' joint states and the gripper widths, the task's instruction and the time. At 1280 x 720 the left wrist camera reads the barrel's scale while a dose is set, about 9 px between 0.1 mL lines; the station and overhead cameras see the syringe, not its scale. It is not given the liquid's state, a dose, a pressure or the evaluator's view; the gateway's entry checks look for those words in every observation (section 18).

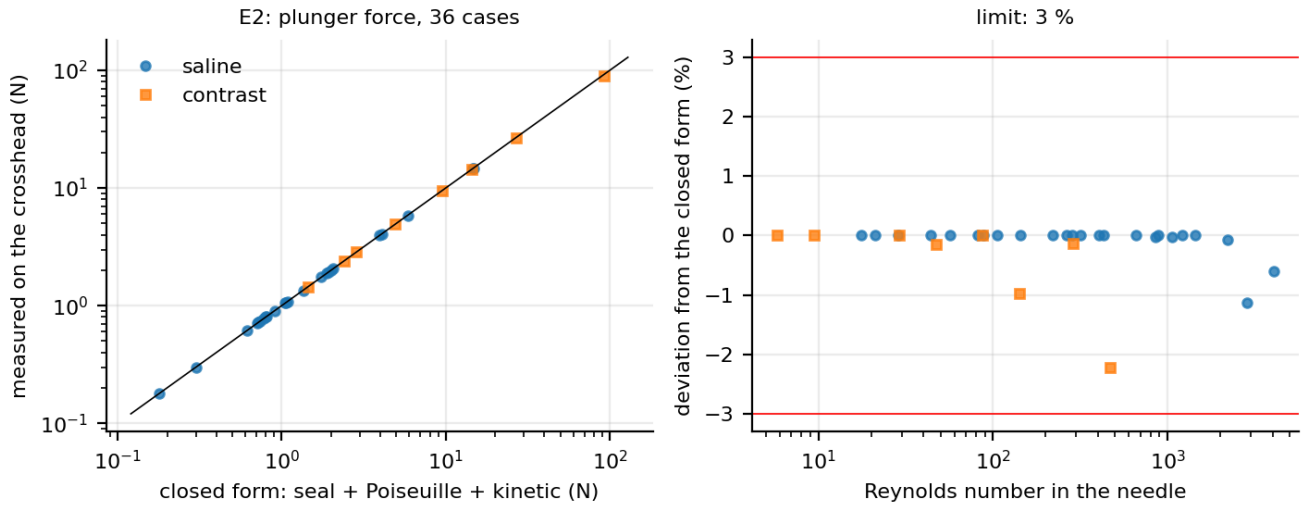
04 / What the plunger pushes against

A lumped liquid network solved implicitly with the rigid world; checked against its own closed form on a testing machine.

A syringe's contents are a short list of numbers, not particles: a chamber pressure p_c, a stack of liquid parcels in order from the nozzle, an amount of gas, one flow Q along the path and the pressure p_t where the path ends (the air, a cap, a vial, the saline lock's line, the pad's tissue). With the plunger's speed v and the bore's area A:

$$C_c \frac{dp_c}{dt} = Av - Q, \qquad L\frac{dQ}{dt} = p_c - p_t - (R + 2k|Q_0|)\,Q - s \tag{1}$$

$R = 128\mu\ell/(\pi d^4)$ is Poiseuille's resistance over the nozzle, the needle or cannula and any line, $k = K\rho/(2a^2)$ the kinetic loss at the narrowest bore, L the inertance and C_c the barrel's, the stopper's and the gas's compliance. Solved over the tick with the plunger's speed left open, the force on the plunger is $F = F_{bias} + c_h v'$. This velocity-dependent force is solved as a bilateral constraint row together with contact and seal friction, with no plunger armature: an armature would be numerical inertia, not measured mass. The stopper's seal is dry friction that rises from the glide force toward the break-loose force with the time it has stood, $F = F_g + (F_b - F_g)(1 - e^{-t_{rest}/20\,s})$, inside ISO 7886-1 Annex G's limits.



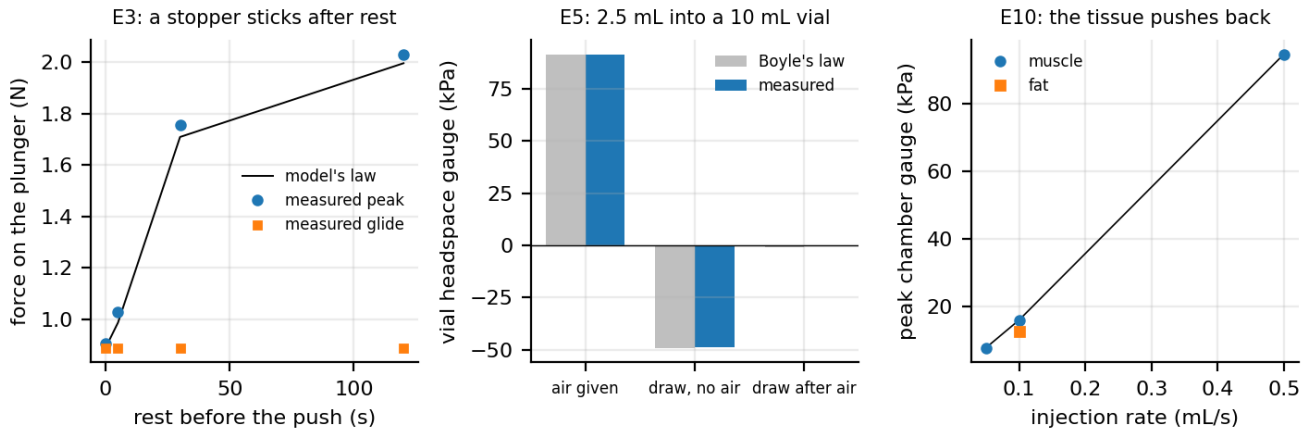
E2: the plunger's force on a testing machine against the closed form, 36 combinations of syringe, needle, liquid and speed.

Against published force measurements (E2b), the simulator is compared needle family by needle family, with an uncertainty envelope. The authored trainer hub has 1 mm of narrow embedded cannula and a wider passage after it. Predicted over published, as geometric means: 1.277 for the short 27 G needle, 1.075 for the 27 G spinal needle and 0.691 for 30 G. These are not a collective pass: the short-needle bias and the underprediction at high flow are errors of the model. The paper does not fully specify the fluid or the manufacturing geometry, so the comparison is bounded over viscosity, bore, embedding and seal assumptions without refitting each row.

measured on the testing machine (E2, E13)	value	limit
cases (syringe, needle, liquid, speed)	36	
worst force against the closed form	2.23 %	3 %
worst liquid out against the plunger's travel times its area	0.00215 mL	
largest change of the force at half the tick	0.00130 %	0.5 %

05 / Air, vials and the line

Which phase leaves first, a vial as a closed vessel, the saline lock's line and its clamp; each against its own prediction.



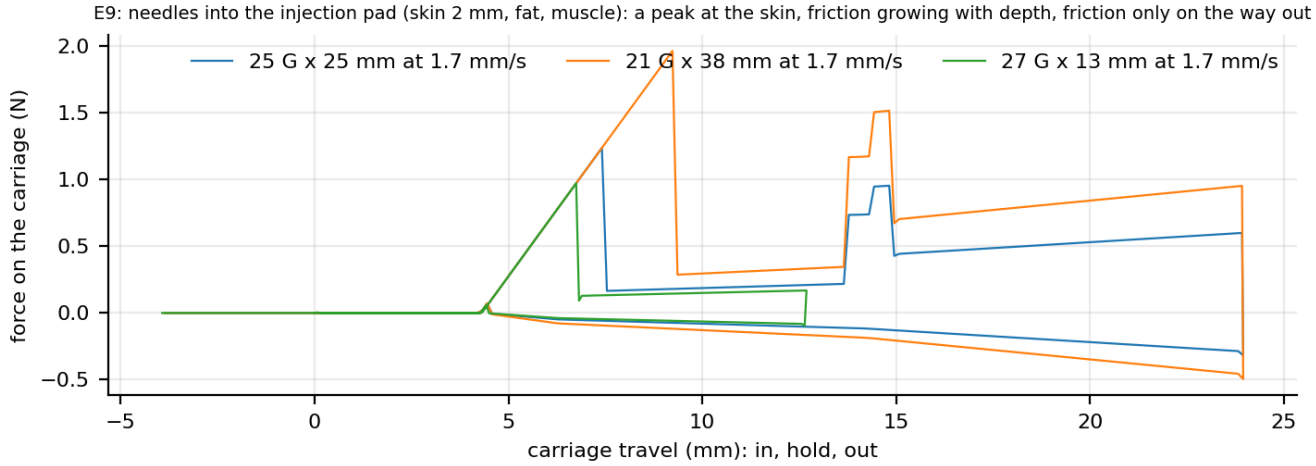
E3, E5 and E10: the stopper's peak force against its rest before the push (the glide stays put); the vial's headspace after 2.5 mL of air is given and after a draw without air, against Boyle's law; the chamber's peak pressure against the injection rate in muscle and in fat.

Gas moves along the barrel toward "up" as the liquid feels it and leaves first only when the bubble is at the nozzle; in the 3 and 10 mL barrels a bubble rises freely, in the 1 mL barrel it stays until the barrel is flicked (E4). A vial holds liquid and a headspace with pV_g constant; the needle's bevel is in the liquid or in the gas by the vial's attitude (E5). The saline lock's line holds 1.1 mL, so medication arrives in the vein only after the saline ahead of it; a shut clamp makes the push a pressure rise and nothing else (E7). The tissue stores what is injected and gives back a share through the track if the needle leaves at once (E10).

	what it checks (a bench, no robot)	measured	result
E3	the stopper's break-loose force grows with rest; a stored stopper is stuck until freed	2 of 2 criteria held; peak 0.905 N after no rest, 2.03 N after 120 s (glide 0.890 N)	**PASS**
E4	air leaves first only with the bubble at the nozzle; the 1 mL bubble stays put until flicked	3 of 3 criteria held	**PASS**
E5	the vial's headspace follows Boyle's law; without air given first the vacuum pulls the plunger back	3 of 3 criteria held	**PASS**
E7	medication arrives after the line's 1.1 mL of saline; a shut clamp is force over area; a pull brings blood	4 of 4 criteria held; ledger error 2.03e-14 mL	**PASS**
E10	the tissue's counter-pressure by layer and rate; leak-back only when the needle leaves at once	2 of 2 criteria held	**PASS**
E11	drops by Tate's law; jets fly with air drag; nothing lost in the ledger	2 of 2 criteria held	**PASS**

06 / Needles, the pad and the Luer lock

A point that dimples, punctures and then slides in its track; a Luer fitting that seals only when it is turned.



E9: the force on the carriage as each needle goes into the pad and out: a peak as the point breaks the skin, a step at each layer, friction that grows with depth, friction only on the way out.

A needle is a hub, a tube and a bead that slides on the tube. On a soft face the point presses a dimple until the force reaches the layer's puncture force, scaled with the needle's diameter (25 G into skin 1.24 N, an 18 G blunt fill needle into a stopper 11.8 N); then the bead is fixed at the entry point and the tube slides through it, with friction that grows with depth and a cutting force on an advancing point only. The bending moment at the hub is compared with the tube's yield moment. A sheath comes off to a pull of 6 N along the needle and is only seated harder by a push (E15). A Luer nozzle goes into a fitting to the start of the thread and no further unless it is turned; turned, it follows the 5 mm lead to the cone's seat, and the joint then holds against a pull (E6).

	what it checks (a bench, no robot)	measured	result
E6	pushed on and not turned is not sealed; turned, it seats, holds 35 N and does not override below 0.15 N m	7 of 7 criteria held	**PASS**
E9	a puncture peak at the skin, friction growing with depth and only friction on the way out; a levered needle is flagged bent	3 of 3 criteria held	**PASS**
E15	a sheath lets go to a pull of 6 N and not to a push of 15 N	pull let go at 6.00 N; push to 15.0 N, still seated	**PASS**
E1	the engine facts the model relies on: subtree masses, `mj_setConst`, checkpoints, the solver's tolerance	every fact held	**PASS**

07 / The robot and what it is given

Two FR3 arms with Franka Hands on a cart; joint targets and gripper commands only; the operator is a script with privileged state.



The station at the handover: set-dose (left) and flush-lock (right).

The arms are the FR3 with the Franka Hand, on the shared runtime's servos (finite effort, the FR3's joint limits, a collision reflex that stops an arm above its torque threshold). The fingertip pads are one normal spring of 23 N/mm a pad on what they hold, whatever the number of contact points; a held 3 mL barrel slid 0.006 mm in the hand over a lift and a joint move (E12). The two Pandas, a single FR3 and the G1 humanoid are also offered (sections 12 and 23).

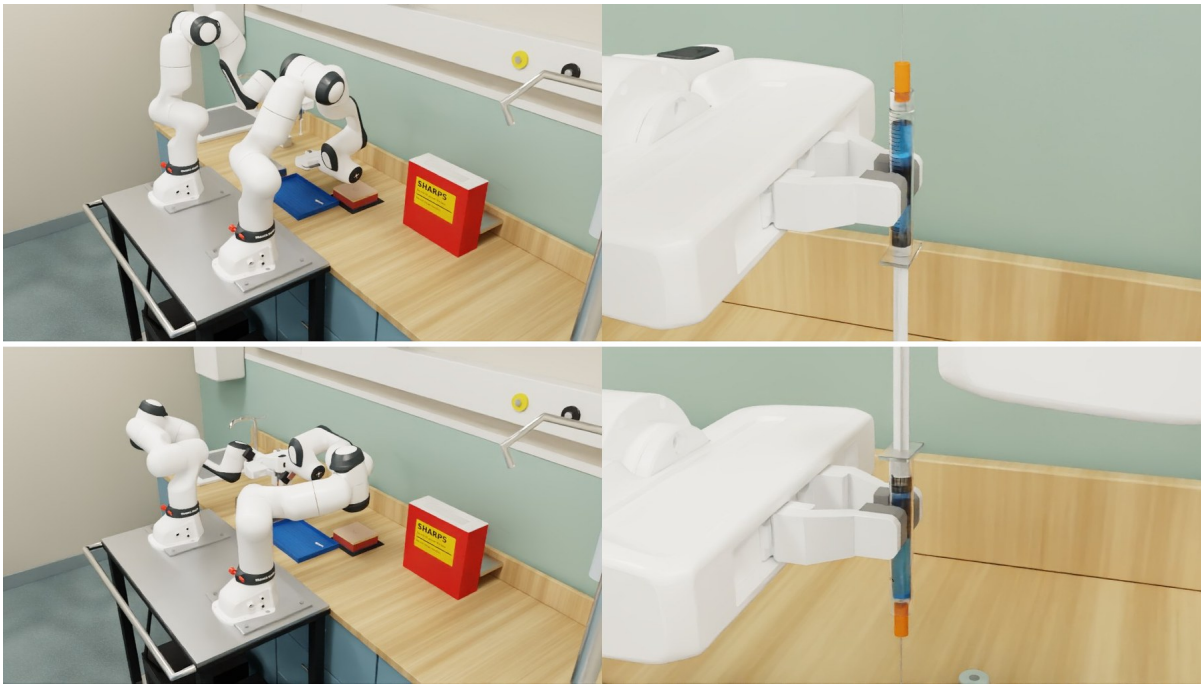
Every demonstration here is driven by a scripted operator. It reads object poses and the liquid's state, which a policy does not get, and plans with the collision engine's clearances; what it sends is joint targets and gripper commands, nothing else: no force on an object, no attachment, no pose written. The footer of every film says so. The robot runs of sections 08 to 17 were recorded on an earlier version of the physics (section 24).

the grasp (E12)	measured	prediction
pad sink at 5, 15, 35 N against force over stiffness	1.00, 1.00, 0.999 (ratio)	1 within 10 %

E12: **PASS**.

08 / set-dose

Sheath off, needle up, the bubble to the nozzle and out, the excess into the basin, 2.5 mL on the scale.



set-dose by the robot, wide shot and close view: the bubble at the nozzle with the needle up (top), the right hand's closed fingers pressing the plunger down to 2.5 mL over the basin (bottom).

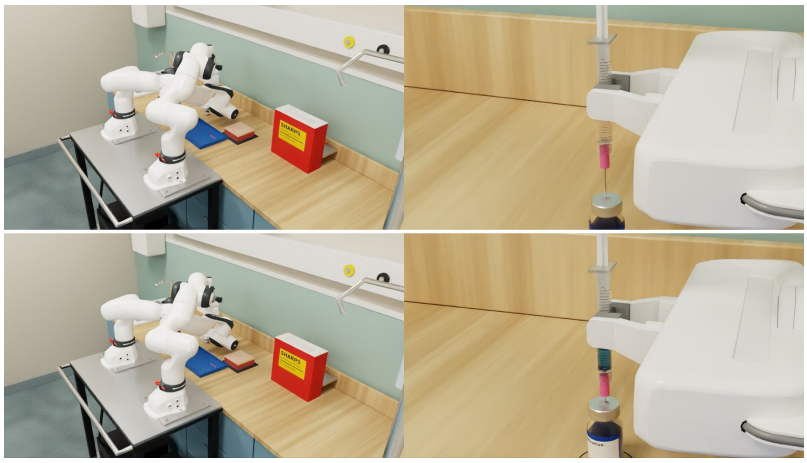
The left hand takes the barrel and lifts it level; the right hand comes at the sheath from its closed end, pulls it off along the needle and lays it on the tray. The left hand turns the syringe needle up and the bubble rises to the nozzle; the right hand's closed fingers press the plunger until the air is out, then, with the needle down over the basin, down to the 2.5 mL line.

One continuous run of 76.0 s of simulated time; the evaluator's verdict **success**, score 1.00; 0 protective stops; no robot link touched the world in 75951 ticks.

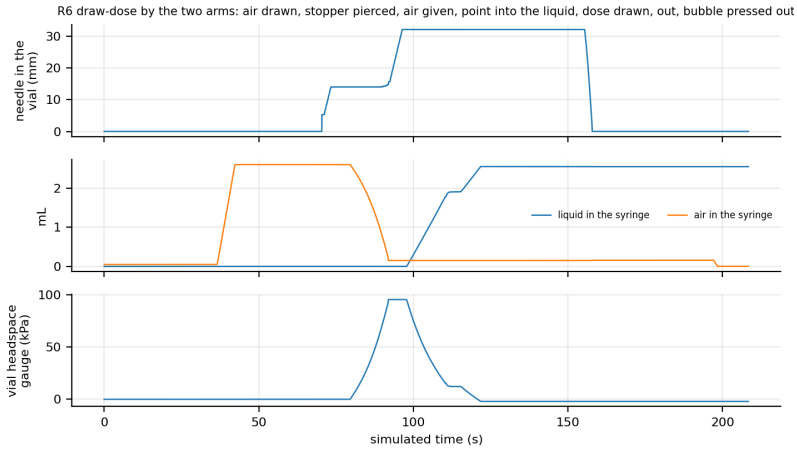
what is judged	measured	limit	held
dose in the syringe (mL)	2.50	2.5 +/- 0.125 mL (ISO 7886-1 Table 1)	yes
air in the syringe (mL)	0.00960	<= 0.05 mL, in a syringe that holds at least half the dose	yes
the 8 conduct criteria	8 held; spilt 0.00630 mL	every one	yes

09 / draw-dose

Air given first, the dose drawn upright and a handover for withdrawal.



draw-dose by the robot: the needle through the stopper as the air goes in (top), the vial's pressure driving the plunger out after the rising hand (bottom).



draw-dose as measured in the run: the needle's depth in the vial, the liquid and air in the syringe, and the vial's headspace rising as the air goes in and falling as the dose comes out.

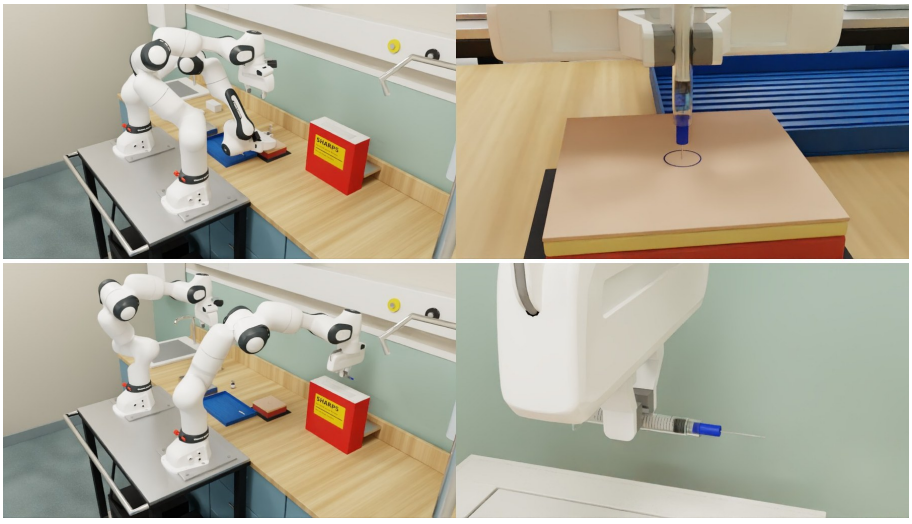
The robot gives air before drawing liquid. The vial's pressure drives the plunger after the rising hand; a handover lets the other hand hold the vial for withdrawal. On the physics as it is, a run of the same operator stopped before the end (section 24).

One continuous run of 208 s of simulated time; the evaluator's verdict **success**, score 1.00; 0 protective stops; no robot link touched the world in 208388 ticks.

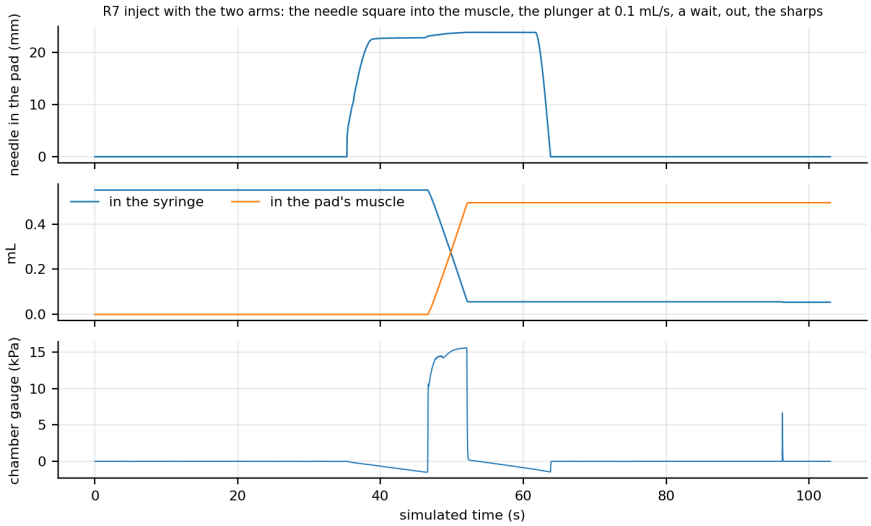
what is judged	measured	limit	held
dose in the syringe (mL)	2.50	2.5 +/- 0.125 mL (ISO 7886-1 Table 1)	yes
air in the syringe (mL)	0.00250	<= 0.05 mL, in a syringe that holds at least half the dose	yes
the vial	tilt cosine 1.0000, moved 4.60 mm, needles in it: none	standing upright on the counter, nothing in it or on it	yes
the vial's headspace (kPa over ambient)	-1.99	headspace within +/- 20 kPa of ambient	yes
the vial's loss beyond the dose (mL)	0.00190	the vial has lost the dose and at most 0.1 mL more	yes
the 8 conduct criteria	8 held; spilt 0.00190 mL	every one	yes

10 / inject

Into the muscle at the ring, square, at the dose's rate, a wait, out along the track, into the sharps.



inject by the robot: the needle driven into the pad's ring at 5 mm/s (top), the used syringe let go over the sharps container's slot (bottom).



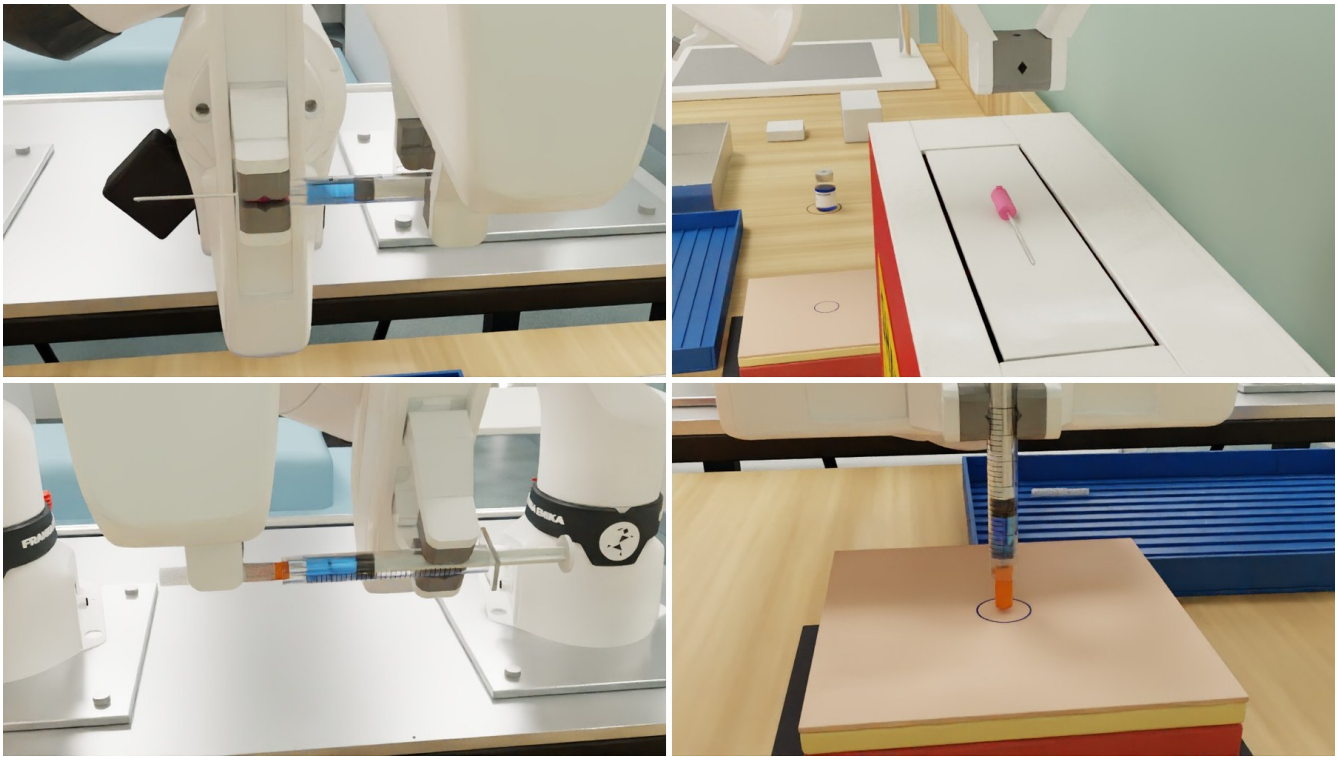
inject as measured in the run: the needle's depth in the pad, the dose leaving the syringe for the muscle, the chamber's pressure.

One continuous run of 103 s of simulated time; the evaluator's verdict **success**, score 1.00; 0 protective stops; no robot link touched the world in 102999 ticks.

what is judged	measured	limit	held
the dose by layer of the pad (mL)	muscle 0.494	>= 90 % of 0.5 mL in the muscle; <= 5 % back out of the track	yes
largest volume in any 2 s (mL)	0.191	<= 0.3 mL in any 2 s (1.5 times 10 s per mL)	yes
air into the patient (mL)	0	<= 0.05 mL of air into the tissue	yes
the needle's path	1.0 mm from the ring's centre, 1.1 deg from square, 23.9 mm deep, levered 0.5 deg	into the 20 mm ring, within 15 deg of square, not levered or withdrawn more than 15 deg off the way in	yes
the sharps container	inside: the syringe, the syringe's needle; outside: nothing; flap 0 deg	every used syringe with its needle inside, flap closed, nothing else in it	yes
the 8 conduct criteria	8 held; spilt 0.00190 mL	every one	yes

11 / vial-to-pad

Draw, needle change, injection and disposal in one continuous run.



vial-to-pad by the robot: the right hand pinching the fill needle's hub while the left unscrews the syringe from it (top left), the needle laid on the sharps container's flap (top right), the 25 G needle turned on (bottom left), the dose into the pad (bottom right).

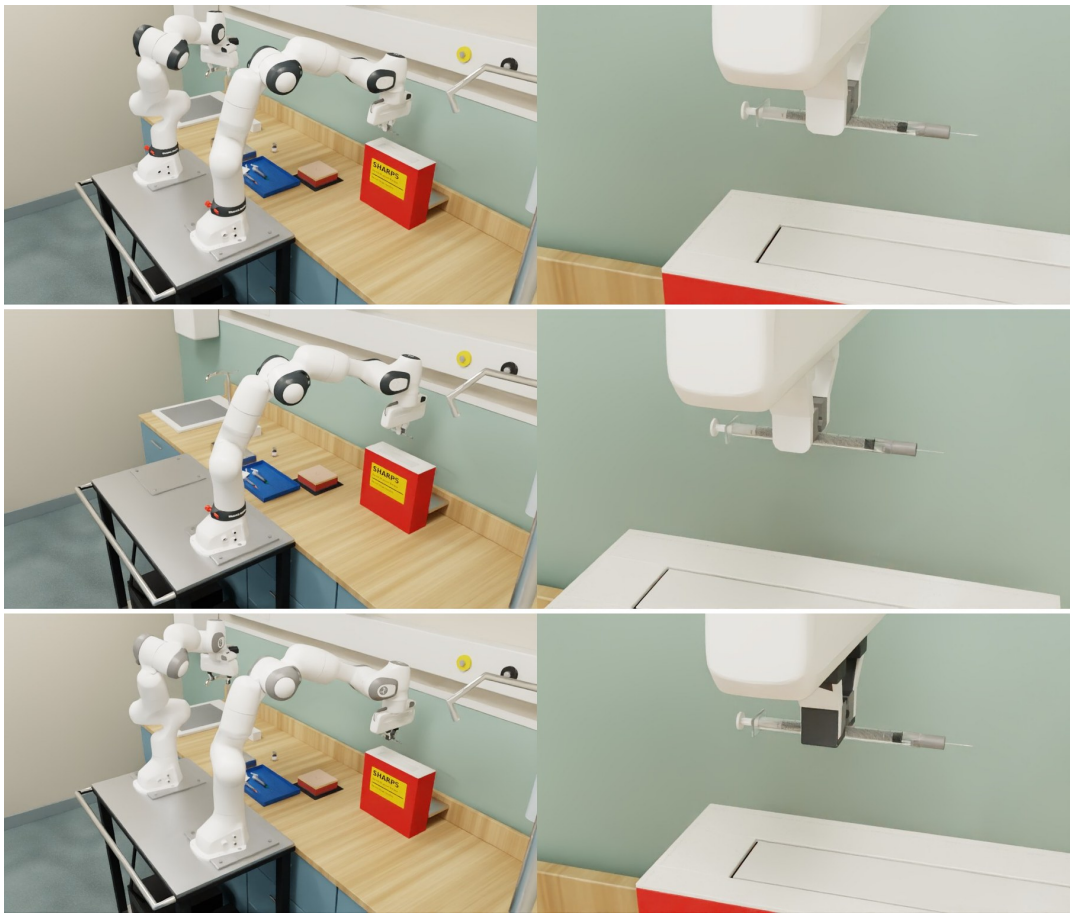
The fill needle comes off in five roll strokes of the left wrist while the right pinches its hub, eased to 3 N once the thread is free; the right hand lays it on the flap and presses the flap down with its closed fingers beside it. The 25 G needle is taken by its sheath from the tray, the nozzle pushed into its hub and turned on in eleven strokes until sealed, and the sheath pulled off along the needle. No handover of the used syringe in the air cleared in this run; the left hand laid it in its channel with its finger grips flat, the bare needle clear of the tray, and the right carried it level over the slot. On the physics as it is, the operator stops during the needle change (section 24).

One continuous run of 405 s of simulated time; the evaluator's verdict **success**, score 1.00; 0 protective stops; no robot link touched the world in 404906 ticks.

what is judged	measured	limit	held
the dose by layer of the pad (mL)	muscle 0.991; surface 6.00e-04	>= 90 % of 1 mL in the muscle; <= 5 % back out of the track	yes
largest volume in any 2 s (mL)	0.193	<= 0.3 mL in any 2 s (1.5 times 10 s per mL)	yes
air into the patient (mL)	0	<= 0.05 mL of air into the tissue	yes
the needle's path	0.4 mm from the ring's centre, 1.0 deg from square, 24.2 mm deep, levered 0.4 deg	into the 20 mm ring, within 15 deg of square, not levered or withdrawn more than 15 deg off the way in, with the 25 G needle	yes
the vial	tilt cosine 1.0000, moved 3.00 mm, needles in it: none	standing upright on the counter, nothing in it or on it	yes
the vial's headspace (kPa over ambient)	-1.56	headspace within +/- 20 kPa of ambient	yes
the vial's loss beyond the dose (mL)	0.00230	the vial has lost the dose and at most 0.1 mL more	yes
the sharps container	inside: the syringe, the fill needle, the 25 G needle; outside: nothing; flap 0 deg	every used syringe with its needle inside, flap closed, nothing else in it	yes
the 8 conduct criteria	8 held; spilt 0.00290 mL	every one	yes

12 / sharps-away, on three robot configurations

Three used syringes through the flap by the two FR3 arms, by a single FR3, and by two Pandas under the same operator.



sharps-away: the two FR3 arms (top), the single FR3 (middle) and the two Pandas (bottom), each with a syringe over the slot.

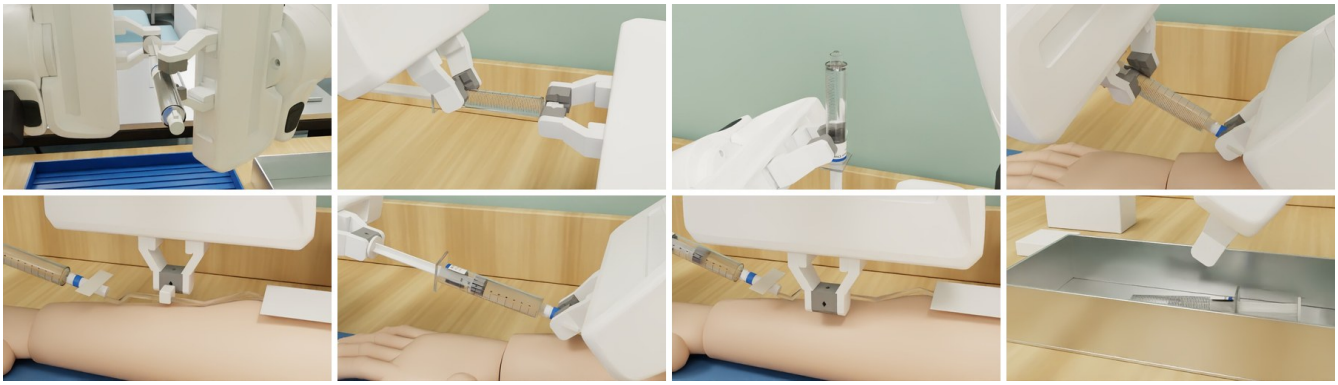
One continuous run of 59.0 s of simulated time; the evaluator's verdict **success**, score 1.00; 0 protective stops; no robot link touched the world in 59011 ticks.

what is judged	measured	limit	held
the sharps container	inside: syringe A (3 mL), syringe B (10 mL), syringe C (1 mL), A's needle, B's needle, C's needle; outside: nothing; flap 0 deg	every used syringe with its needle inside, flap closed, nothing else in it	yes
the 8 conduct criteria	8 held; spilt 0 mL	every one	yes

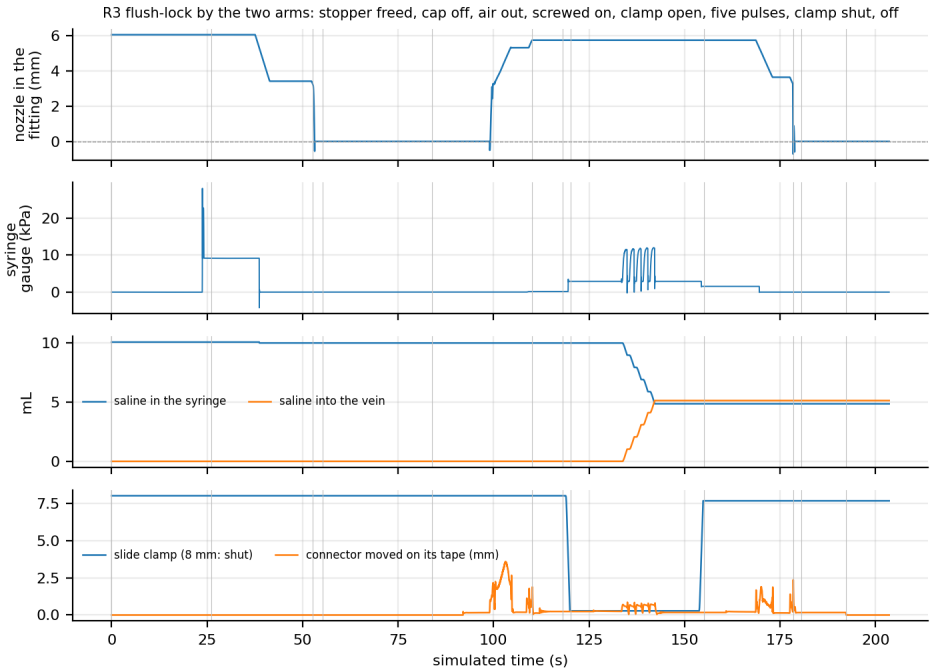
The single FR3 and the two Pandas pass sharps-away with the same operator (58 and 61 s). On the Pandas, set-dose ended with two protective stops and 2.80 mL left in the syringe, and inject found no clear way for the left arm to the pad: the operators were written for the FR3 arms' reach, and the lock and vial tasks have not been tried on the Pandas. The catalogue offers the Pandas and the single FR3 with that said (`robot_setup.py`).

13 / flush-lock

Stopper freed, cap unscrewed, air out, screwed onto the connector, clamp open, five pulses, clamp shut, unscrewed, into the dish.



flush-lock by the robot, close views of eight steps: the stored stopper pressed free against the cap, the cap unscrewed by the right wrist, the bubble pressed out, the syringe turned onto the connector, the clamp pushed open, a 1 mL pulse, the clamp pushed shut, the syringe into the dish.



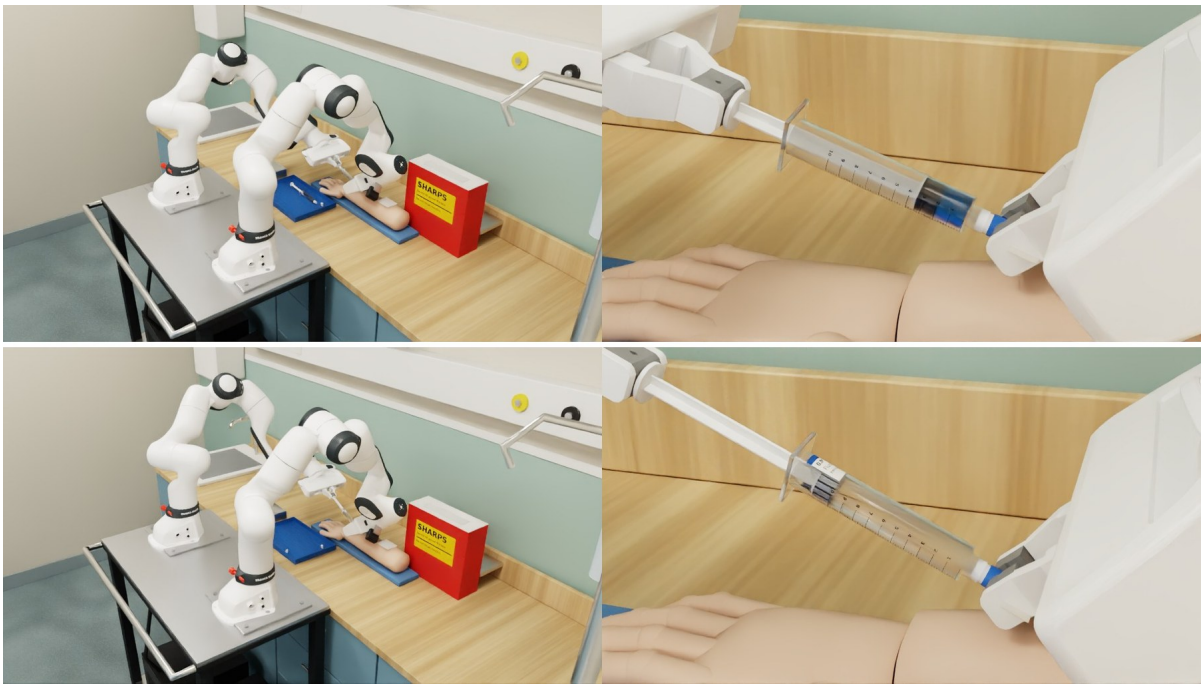
flush-lock as measured in the run: the nozzle in the cap's or the connector's fitting, the syringe's pressure (the stopper's break-free, the five pulses), the saline, the clamp and the connector's movement on its tape.

One continuous run of 204 s of simulated time; the evaluator's verdict **success**, score 1.00; 0 protective stops; no robot link touched the world in 203569 ticks.

what is judged	measured	limit	held
the slide clamp at the end (mm across)	7.66	slide clamp closed	yes
anything left on the connector	none	nothing attached to the connector	yes
the connector's largest shift (mm)	3.57	<= 10 mm at any time	yes
the line's peak gauge pressure (kPa)	12.3	<= 300 kPa at any time	yes
air into the patient (mL)	0	<= 0.05 mL of air into the vein	yes
saline into the vein (mL)	5.13	>= 95 % of 5 mL of saline into the vein	yes
the 8 conduct criteria	8 held; spilt 0.0156 mL	every one	yes

14 / slow-push

Four millilitres through the lock at the dose's rate over two minutes, then the flush with its first millilitre at the same rate.



slow-push by the robot: the left hand pushing the dose at 0.033 mL/s with the right holding the connector (top), the flush syringe on the lock after the dose (bottom).

The medication goes in behind the 1.1 mL of saline already in the line, so the vein receives saline first; the flush carries the last of the dose in, which is why 5 % to 95 % of the dose arrives over 312 s. The second syringe lies in the tray's back channel, where the FR3's wrist cannot take it with the hold the later steps need; the operator lays it in the front channel the first syringe left free and takes it again.

One continuous run of 516 s of simulated time; the evaluator's verdict **success**, score 1.00; 0 protective stops; no robot link touched the world in 516417 ticks.

what is judged	measured	limit	held
the slide clamp at the end (mm across)	7.63	slide clamp closed	yes
anything left on the connector	none	nothing attached to the connector	yes
the connector's largest shift (mm)	3.61	<= 10 mm at any time	yes
the line's peak gauge pressure (kPa)	23.4	<= 300 kPa at any time	yes
air into the patient (mL)	0	<= 0.05 mL of air into the vein	yes
medication into the vein (mL)	3.96	>= 95 % of 4 mL of medication into the vein	yes
saline into the vein (mL)	5.10	>= 95 % of 5 mL of saline into the vein	yes
largest dose into the vein in any 15 s (mL)	0.714	<= 0.750 mL of medication into the vein in any 15 s	yes
5 % to 95 % of the dose into the vein (s)	312	5 % to 95 % of the dose in not less than 86 s	yes
the 8 conduct criteria	8 held; spilt 0.0420 mL	every one	yes

15 / fast-bolus

Ten millilitres of viscous contrast in six seconds through a 20 G cannula, then the flush.



fast-bolus by the robot: the bolus pushed at 1.6 mL/s, the right hand holding the connector with the joint solution that bears the push best.

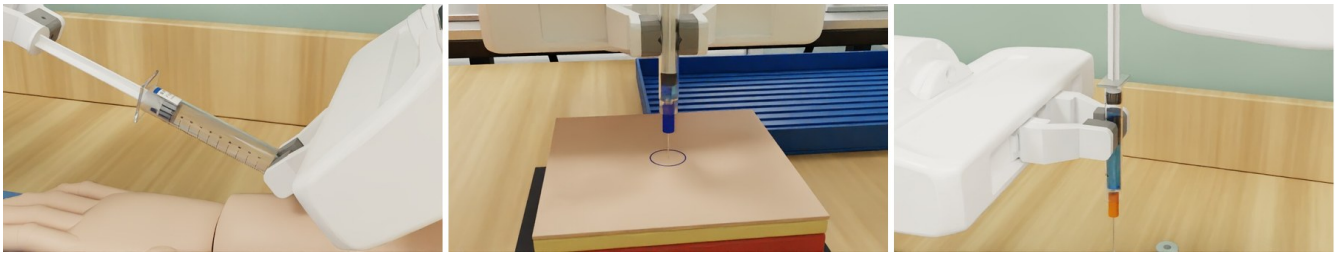
Contrast at 11.6 mPa s through the cannula needs 16 to 20 N on the plunger at 1.6 mL/s; that force and its reaction on the connector act about each arm's first joint at about half a metre, 8 to 10 N m against the FR3's 10 N m holding threshold. The operator pushes with the joint solutions that bear it best; a policy that pushes with another posture will stop the arms at a lower rate.

One continuous run of 356 s of simulated time; the evaluator's verdict **success**, score 1.00; 0 protective stops; no robot link touched the world in 355666 ticks.

what is judged	measured	limit	held
the slide clamp at the end (mm across)	7.63	slide clamp closed	yes
anything left on the connector	none	nothing attached to the connector	yes
the connector's largest shift (mm)	5.24	<= 10 mm at any time	yes
the line's peak gauge pressure (kPa)	144	<= 300 kPa at any time	yes
air into the patient (mL)	0	<= 0.05 mL of air into the vein	yes
medication into the vein (mL)	9.99	>= 95 % of 10 mL of contrast into the vein	yes
saline into the vein (mL)	5.17	>= 95 % of 5 mL of saline into the vein	yes
5 % to 95 % of the contrast out of the syringe (s)	6.30	5 % to 95 % of the dose out of the syringe into the line within 7.2 s	yes
the 8 conduct criteria	8 held; spilt 0.0392 mL	every one	yes

16 / When it goes wrong

Each deviation done by the robot, and the criterion it fails on; one predicted variation that passes.



Negatives by the robot, close views: the left hand pressing the plunger against the shut slide clamp (left; the line's pressure rises, nothing reaches the vein), the needle stopped 9 mm in, in the fat, with the dose going there (middle), the dose set needle down with the bubble still in the barrel (right).

run (robot unless a phantom-hand bench)	the deviation is about	success	score	criteria not held	as predicted
set-dose: air pushed out needle down	air in the syringe (mL)	no	0	dose in the syringe (mL), air in the syringe (mL)	yes
inject: needle stopped in the fat	the dose by layer of the pad (mL)	no	0.667	the dose by layer of the pad (mL)	yes
inject: no wait before the needle comes out	(a predicted variation)	yes	1.00	none	yes
inject: needle levered	needles loaded past their yield	no	0.900	needles loaded past their yield	yes
sharps-away: bare needle on the counter	what a bare needle or nozzle touched that it may not	no	0.875	what a bare needle or nozzle touched that it may not	yes
flush-lock: clamp left shut	saline into the vein (mL)	no	0	protective stops of an arm, what a bare needle or nozzle touched that it may not, the connector's largest shift (mm), the line's peak gauge pressure (kPa), saline into the vein (mL)	yes
slow-push: dose pushed in 5 s	largest dose into the vein in any 15 s (mL)	no	0.929	largest dose into the vein in any 15 s (mL)	yes
draw-dose: no air given to the vial	dose in the syringe (mL)	no	0	dose in the syringe (mL), air in the syringe (mL), the vial	yes
flush-lock, phantom hands: pushed on, not turned	saline into the vein (mL)	no	0	liquid outside the basin, the vein and the tissue (mL), saline into the vein (mL)	yes
flush-lock, phantom hands: clamp left shut	saline into the vein (mL)	no	0	the line's peak gauge pressure (kPa), saline into the vein (mL)	yes
slow-push, phantom hands: dose pushed in 5 s	largest dose into the vein in any 15 s (mL)	no	0.619	largest dose into the vein in any 15 s (mL), 5 % to 95 % of the dose into the vein (s)	yes
draw-dose, phantom hands: no air given to the vial	the vial's headspace (kPa over ambient)	no	0.909	the vial's headspace (kPa over ambient)	yes

Each negative is the task's own operator with one thing changed, run from the same start state; it is judged by the same evaluator and fails on the criterion its deviation is about, and on no criterion it is not about except where the deviation causes it (a shut clamp also raises the line's pressure). The negatives are filmed at the stretch where they go wrong (section 19).

17 / Evaluator checks, repeated runs and variants

The evaluator against doing nothing, every demonstration repeated in fresh processes, and the variants.

Doing nothing: 25 task and variant pairs left alone from their start states; every score after 2.5 s: 0 (**PASS**).

demonstration	runs in their own processes	success	simulated time (s)	criteria whose values differ
set-dose	3	yes, yes, yes	76.0, 76.0, 76.0	none
inject	3	yes, yes, yes	103, 103, 103	none
sharps-away	3	yes, yes, yes	59.0, 59.0, 59.0	none
flush-lock	3	yes, yes, yes	204, 204, 204	none
draw-dose	0	none	none	none
slow-push	3	yes, yes, yes	516, 516, 516	none
fast-bolus	3	yes, yes, yes	356, 356, 356	none
vial-to-pad	0	none	none	none

S4: **FAIL**: draw-dose and vial-to-pad were not repeated.

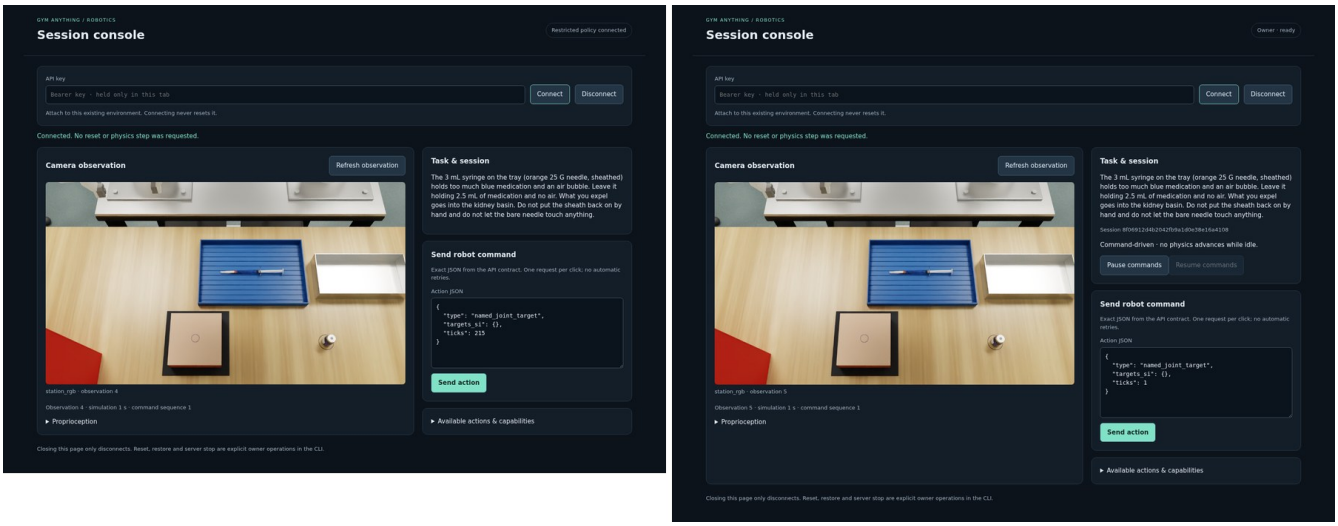
The variants (an upright arm model, a subcutaneous dose, a half vial, a vial under vacuum, a vented adapter, a blocked cannula, a clamp found open, 18 G and 22 G cannulas, the short push) are each done by the same operator from their start states; a variant the operator was not written for is the operator measured against it, and the table says which answered as the evaluator expects and where the operator could go no further.

The robot on every variant: 17 runs, 13 done to the end by the operator, 9 passed by the evaluator.

task	variant	what is different	outcome
draw-dose	adapter	draw-dose through a vented needle-free vial adapter instead of the blunt fill needle	the operator stopped: ValueError: this operator draws through a fill needle; the syringe here has none (the vial adapter's variant): no plan for it
flush-lock	blocked	flush-lock with a cannula that is blocked (do not force it)	failure, score 0; not held: protective stops of an arm, the connector's largest shift (mm), the line's peak gauge pressure (kPa), line_not_forced
flush-lock	clamp-open	flush-lock / slow-push / fast-bolus with the slide clamp found open	success, score 1.00
slow-push	clamp-open	flush-lock / slow-push / fast-bolus with the slide clamp found open	success, score 1.00
fast-bolus	clamp-open	flush-lock / slow-push / fast-bolus with the slide clamp found open	success, score 1.00
flush-lock	g18	the iv-lock tasks with an 18 G cannula	success, score 1.00
slow-push	g18	the iv-lock tasks with an 18 G cannula	success, score 1.00
fast-bolus	g18	the iv-lock tasks with an 18 G cannula	success, score 1.00
flush-lock	g22	the iv-lock tasks with a 22 G cannula	success, score 1.00
slow-push	g22	the iv-lock tasks with a 22 G cannula	success, score 1.00
draw-dose	half-vial	draw-dose / vial-to-pad from a vial that holds 5 mL	failure, score 0; not held: dose in the syringe (mL), air in the syringe (mL)
vial-to-pad	half-vial	draw-dose / vial-to-pad from a vial that holds 5 mL	the operator stopped: ValueError: left: the 25 G needle is not on (depth 3.34 mm, sealed False)
slow-push	short	slow-push as 1 mL over one minute	failure, score 0.929; not held: protective stops of an arm
inject	subcut	inject as a subcutaneous dose: 0.3 mL from a 1 mL syringe through a 27 G x 13 mm needle	success, score 1.00
inject	upright	inject / vial-to-pad with the pad strapped to an upright upper-arm model (its skin vertical, facing the robot)	the operator stopped: ValueError: IK unreachable (1.06 mm, 0.96 deg); no action executed
vial-to-pad	upright	inject / vial-to-pad with the pad strapped to an upright upper-arm model (its skin vertical, facing the robot)	the operator stopped: ValueError: left: no clear way back to its start pose (scene -38 mm, other arm 50 mm)
draw-dose	vacuum-vial	draw-dose from a vial left 25 kPa under ambient by an earlier draw	failure, score 0.500; not held: dose in the syringe (mL)

18 / The interface and its runtime

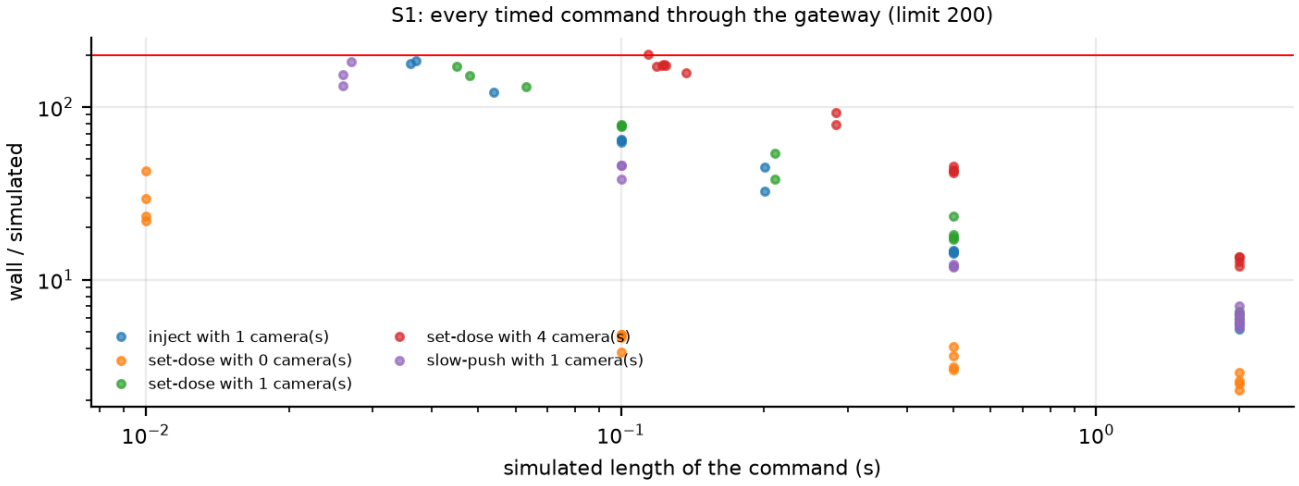
The shared runtime's CLI, HTTP gateway and dashboard; target: at most 200 times real time for every command at the published floor.



The session dashboard as the policy (left, restricted key) and the owner see it.

With the default software camera at 1280 x 720, nine HTTP commands ran at most 31.66 times real time, startup 16.3 s separately; a second session restarted its renderer at the published floor of 0.144 s and ran at 47.89 times real time, PNG delivery included. Four- and five-camera sessions and the Isaac renderer's speed are experimental.

Sessions with the Isaac renderer as the policy camera, measured on an earlier version of the code (the four-camera session exceeded the limit):



S1: every timed command's wall time over its simulated length in the Isaac sessions below (the red line is the limit of 200).

live session	timed commands	worst wall/sim	median	graphics card in use	verdict
inject with 1 camera(s)	16	184	23.7	99 %	PASS
set-dose with 0 camera(s)	16	42.5	3.95	100 %	PASS
set-dose with 1 camera(s)	16	171	30.9	99 %	PASS
set-dose with 4 camera(s)	16	202	61.8	100 %	FAIL
slow-push with 1 camera(s)	16	184	11.9	99 %	PASS

Entry points: **PASS**; refusals checked: one arm for a two-handed task, a misspelt renderer option, 100000 samples a pixel, an unknown task.

19 / Films

Every task film is one continuous run of the robot, rendered from its recording at 25 frames a second; a wide shot of the bay and a close view of the work.

A film is rendered from the run's recording (the poses of every body and the liquid's state at each frame), not from the physics again. The task films use the Isaac renderer with liquid meshes; the component films of section 21 use a section-view replay with translucent contacts and containers, and the sensor films of section 22 use the default policy camera with close framing. In a task film the left tile is the whole bay; the right is a close view chosen per step from three cameras by rendering the recorded moments from each: from the front, from the wall looking back so that no arm comes in front, or from the counter's end. It follows the syringe, or what the free hand works on (a sheath or a cap laid in the tray, the slide clamp, the fill needle into the sharps), and holds at the sharps container's slot once a syringe is inside. The footer says what the robot is doing (the latest step in the run's log) and what the run measured. Negatives are filmed at the stretch where they go wrong, and every robot film ends on a card with the evaluator's verdict. Event clips shoot the decisive moments again from the recording with a camera aimed at the thing judged (the scale while the dose is set, the shut clamp, the needle's point).

20 / Rest, drops, abuse and a spinning needle

Start states and variants left alone, loose things dropped and the robot's commands misused; a spinning needle and a squirting syringe.

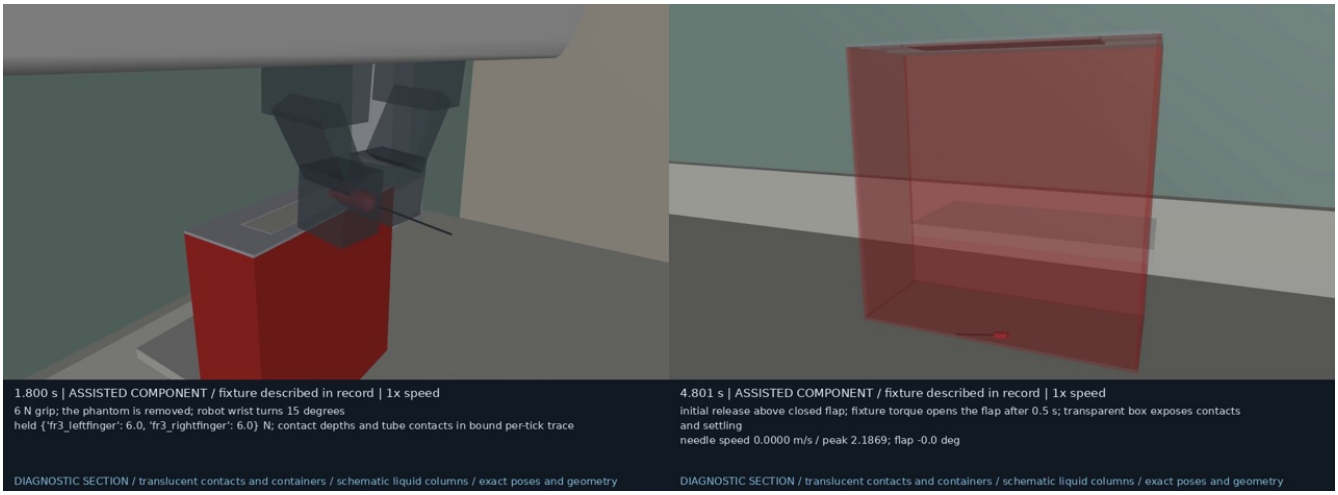
check	result	notes
start states and variants left alone 60 s with the robot present (E0)	25 of 25 at rest	fails: none
every loose thing dropped from 0.2 m, every task (E0)	8 of 8 tasks	37 drops
abuse through the robot's own commands (S2): sweep, slam, squeeze, stab, ram, shake, fling	7 of 7 without a fault	the ledger closed in every one

A spinning needle and the integrator (E14). Pressed by a closed finger onto the sharps flap it lay on, the 0.4 g fill needle went in, rolled on the container's floor spinning at 540 rad/s about its axis and, free, went into a tumble until the world's speed check stopped the run at 16.5 m/s. MuJoCo 3.11's implicitfast integrator steps the gyroscopic term of a free tree of more than one body as Euler does, and the needle is three bodies; its wobble grew 1.136 times a tick, as Euler predicts at that spin, while every rule of the world changed its momentum by exactly zero. The world therefore steps with the implicit integrator. The E14 regression record does not show a pass: its result says `passed=false`, `inside=false`, with zero watched peak speed. **The squirt.** A used syringe landing thumb rest first with 0.1 microlitres of air at its nozzle let 1.84 microlitres of liquid out at the gas's flow rate, a 20 m/s drop that reached the ceiling with 27 times the work the plunger had done, so the chamber uses the gas's law only for a flow the bubble can carry. These flow checks were recorded on an earlier version of the physics; the drop study of section 21 is of the physics as it is.

21 / Component checks of the mechanics

Assisted benches of the needle, the grip, the vial's wall, the Luer lock and the line; the assistance is stated in each record.

The needle has grounded elastic deflection in series with the trainer track. Implicit pad forces keep the 23 kN/m stiffness on gram-scale parts, and implicit hydraulic loading needs no artificial plunger inertia. Eight thin-needle insertions and one surface-contact case were recorded, with six grips at two offsets; a shallow 27 G case punctures and reaches 2.54019 mm before drift. Three independent cantilever-force checks and four needle sizes at two spin-test timesteps bound the reduced model. The 0.5 ms integration tick is a numerical requirement; a captured track uses freshly computed native poses to avoid a false entry impulse.



Component views from recordings: actual fingers gripping an offset bare hub at 1.8 s (left), and the released needle at rest inside the transparent diagnostic sharps box at 4.8 s (right). Transparency exposes contact and settling; the physical geometry and poses are unchanged.

A needle tilted 30 degrees in the vial meets the glass: its tip reaches at most 10.4027 mm from the axis, inside the 11 mm inner radius, with a 0.635 mm tube radius and about 0.038 mm of contact penetration, and the glass pushes back with at most 17.533 N. Forced further, the needle is damaged rather than escaping through the wall. The normal axial draw obtains 2.54904 mL. The closed hydraulic path for outside outlets is independent of the mechanical boundary.

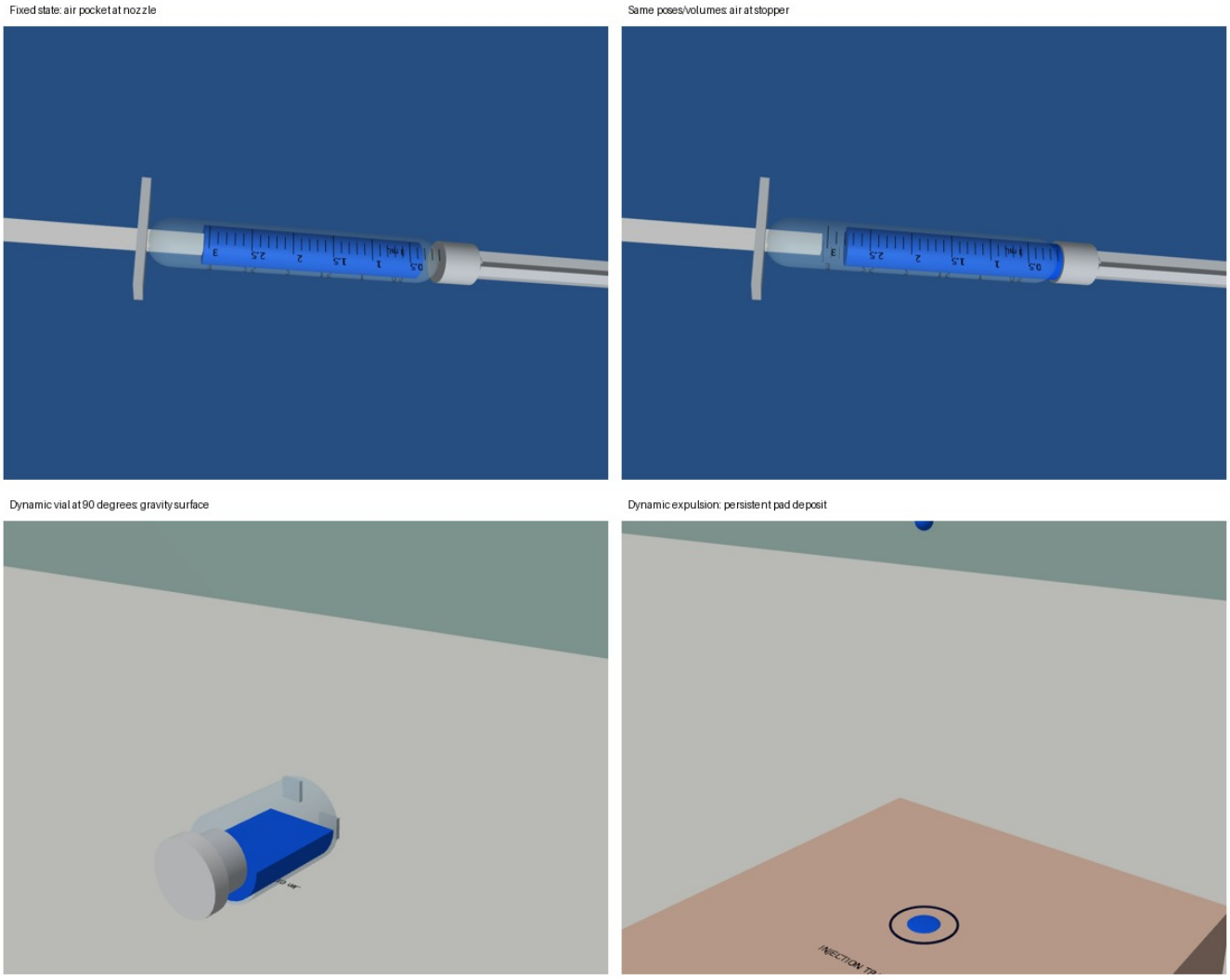
Luer engagement needs 0.2 mm of actual insertion and geometric clearance of the 4.29 mm mouth around the 3.96 mm tip. Parts that are apart stay apart; an offset rim encounter gives 0.04036 N of resistance. An aligned replacement becomes threaded and sealed at 5.8958 mm depth. The fill needle unthreads, comes free at 1.405 s and separates. A force fixture of finite inertia holds the hub; its 50 g of translational and 1e-4 kg m2 of rotational inertia are assistance, not masses of the parts.

Medication crosses the line, first filled with saline, at 0.03174 mL/s and first arrives at 33.87 s; contrast crosses at 1.59741 mL/s and arrives at 0.81 s. The syringe's peak gauge pressures are 9.600 and 76.120 kPa. These are prepared delivery fixtures, not complete slow-push or fast-bolus task passes.

A drop study with no plunger armature covers nine combinations of timestep and extra inertia, expelled liquid included. Its records begin before the release and include contact, speed, warning and fault, and settling observations.

The component films show grip, sheath and dose handling, a clamped vial draw, injection into the pad, a phantom-hand Luer, clamp and flush sequence, needle and syringe release, a hands-off rest and a bounded impact case. Their line-content strips use the saved parcel queue and are schematic. These diagnostic views are not the policy cameras.

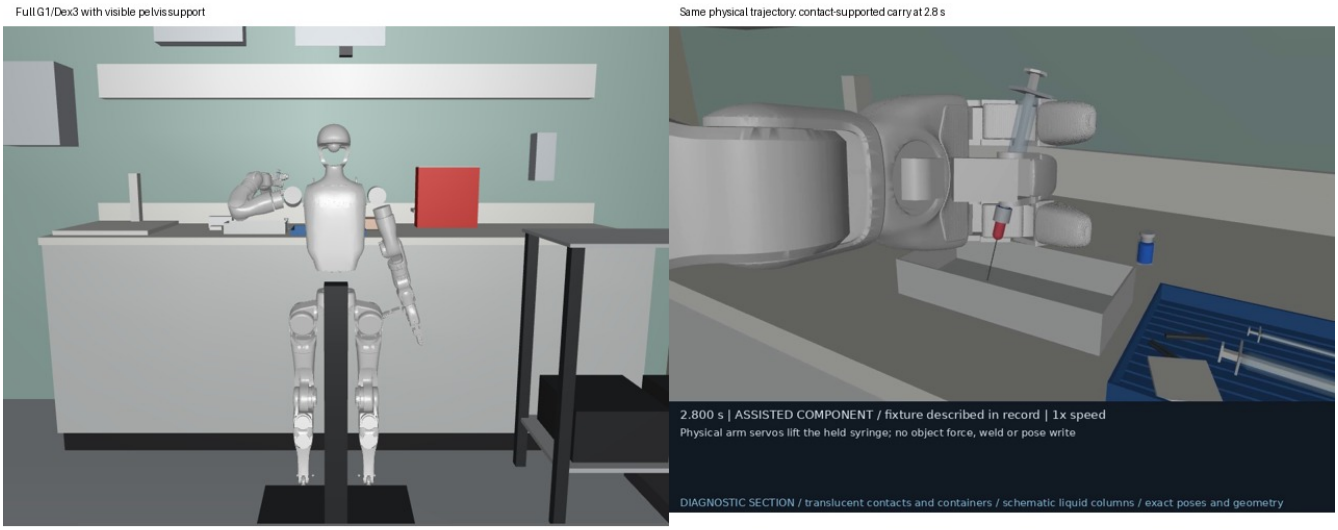
22 / The default policy camera



Default software sensor: same-pose gas-location counterfactuals, a dynamically tilted vial and a persistent pad deposit. These are reduced physical cues, not outcome overlays.

The default policy camera is a persistent CPU software renderer (EGL). It draws the actual poses and rigid geometry, graduation ticks, the gas where purging puts it and the gravity surface vial aspiration uses, the printed injection ring, capacity numerals and equipment labels, clock hands driven by simulation time, and deposits fixed to a body after airborne liquid lands. Three short dynamic films hold 61, 41 and 41 corresponding saved states and sensor frames; the liquid is conserved to 1.06e-15 mL.

23 / The G1 humanoid



Full shared G1/Dex3 at the counter with a visible pelvis stand; close view of the same saved physical trajectory. The stand is explicit assistance, not a balancing controller.

The pinned G1 has 43 body and hand joints, both native Dex3 hands and shared controller profiles. A rigid pelvis support stabilises the full model; the Franka cart is moved aside. The CLI exposes named-joint control in radians. Three live camera commands ran at most 55.38 times real time, with 44.6 s of startup separately. Only sharps-away is offered; Cartesian, parallel-gripper, walking and bimanual clinical control are not qualified.

A prepared syringe settles 18.688 mm when its temporary support is removed. It then moves 0.163 mm during the measured half-second hold and rises 25.381 mm with the arm. Relative motion in the hand over the two-second hold and lift is at most 0.547 mm. Contact is present throughout 200 sampled states, with thumb and index both carrying load in 198. No object fixture acts during this interval. Opening does not fully release: the syringe descends only 13.576 mm. The trial as a whole failed; only the contact-supported hold and lift are credited. No full humanoid disposal task is claimed.

24 / What is not right

The limits that remain, with their numbers where they were measured.

Some choices of the model follow from faults found in testing: parts that would rock between flat pads collide as capsules, drops fly with air drag, a needle sheath lets go to a pull and not to a push (E15), the world steps with the implicit integrator and the chamber uses the gas's law only for a flow the bubble can carry (section 20).

- *The robot runs are of an earlier version of the physics.* The runs of sections 08 to 17 were recorded before the needle's elastic deflection, the implicit pad and hydraulic forces, the vial's glass boundary and the Luer engagement geometry of section 21, and the robot and the variants have not been run again since. On the physics as it is, the draw-dose run stopped before the end and the vial-to-pad operator stops during the needle change. The phantom-hand draw-dose bench (A6) faulted; a clamped vial draw is shown separately.
- *Force.* Against published measurements the force model reads 1.277 times high for the short 27 G needle and 0.691 times low for 30 G (section 04); there is no aggregate force-accuracy pass.
- *The needle.* It deflects elastically but takes no plastic shape past yield: a needle loaded past its yield moment is a terminal flag. Capsule contact envelopes approximate the molded parts.
- *The pad.* It is a synthetic trainer whose storage, permeability and leakage are authored and varied jointly; nothing here is transferred to a patient. Waiting before the needle comes out is not needed for success (section 16).
- *Impacts.* The drop study's energy budget leaves out exact recoil and every contact's stored energy.
- *The drawn liquid.* Air pockets are volume-equivalent shapes and the vial is a clipped cavity; there is no CFD, sloshing or detailed wetting.
- *Speed.* The default software camera is measured on the code as it is (section 18); four- and five-camera sessions and the Isaac renderer are experimental.
- *Other robots.* The operators were written for the FR3 arms' reach (section 12). The G1 stands on a rigid pelvis support, is offered only for sharps-away, and its full trial failed (section 23).

25 / Reproduction

The environment is the package `syringe_handling`, started and driven with the `gar` command.

```
gar inspect syringe-handling ; gar tasks inspect syringe-handling ; gar doctor syringe-handling
gar run syringe-handling --task set-dose --port 0          # then gar status / observe / action / verify / stop
gar run syringe-handling --task sharps-away --robot g1 --port 0
```

26 / References

Opened and read for the research.

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Engine: Newton 1.5.0 with MuJoCo 3.11 (google-deepmind/mujoco, the address in its installed metadata). Robot models: Newton's franka_emika_panda bundle from franka_description at d9841cf, Apache-2.0 (`ASSETS.md`).